01

Mission Control

Mission, release targets, delivery approach and the shared Definition of Done.

BUSINESS QUESTION

What transaction volumes should FRO expect at Golden, Westminster and Parker, and how does uncertainty affect each proposed investment?

INTERNAL RELEASE TARGET

Tue, Nov 3, 2026, 12:00 PM

ASSIGNMENT DEADLINE

Tue, Nov 3, 2026, 5:30 PM

DELIVERY APPROACH

Sprint 0
Project definition, architecture, backlog refinement and Sprint planning

→

Sprint 1
Sunday delivery increment

→

Sprint 2
Monday delivery increment

→

Tuesday
Release buffer and defect correction

DEFINITION OF DONE

1 Stakeholder requirement satisfied

2 Acceptance criteria satisfied

3 Technical implementation verified where applicable

4 QA passed

5 Numbers and evidence reconcile

6 Dale understands the relevant midterm concept

7 Product Owner accepts the story

8 Recovery state preserved when work materially changes the product

02

Starfleet Delivery Teams

Command roles and the three technical workstreams.

Dale
Product Owner / Business Analyst

JARVIS
Scrum Master / QA / Learning Coach

TECHNICAL WORKSTREAMS

Team Defiant
Tech Lead: Dume
Domain: Excel and Modeling

Responsibilities
Statsbook, Excel formulas, regression workbook, profitability model, DCF integration, dashboard

Team Voyager
Tech Lead: Friday
Domain: Python and Development

Responsibilities
Python, Jupyter, pandas, seaborn, EDA, pair-programming support

Team Enterprise
Tech Lead: Ultron
Domain: Word and Communication

Responsibilities
Analysis document, executive communication, formatting, final document quality

Miro Shared cross-team visual planning and architecture workspace. A tool, not a separate team or persona.

03

Epics

Product Backlog: the persistent requirements and traceability view. Placement here is not approval for development.

E1
Excel Statsbook

A4-00 | Establish Assignment 4 Development Workspace
Epic: E1 Excel Statsbook
Team / owner: Defiant / Dume
Story points: TBD during refinement
Dependencies: PRE-A4-01, PRE-A4-02
Refinement: REFINEMENT REQUIRED
Kanban status: BACKLOG

User story
As the Product Owner, I want a controlled Assignment 4 development workspace so that required analytical work can be completed without compromising the verified baseline.

Acceptance criteria
AC1. Assignment 4 development uses an identifiable working baseline.
AC2. Required source materials are available to the development workflow.
AC3. Existing verified functionality is preserved.
AC4. Recovery state is preserved before material model changes.
AC5. Workspace supports the required Excel/modeling development activities.
AC6. No unnecessary architecture or functionality is introduced beyond Assignment 4 requirements.

A4-01 | Prepare Statsbook Data
Epic: E1 Excel Statsbook
Team / owner: Defiant / Dume
Story points: 2
Dependencies: verified project baseline
Refinement: ACCEPTED
Kanban status: DONE

User story
As an analyst, I want the Assignment 4 Statsbook data prepared for analysis so that subsequent categorical, numerical, regression, and forecasting work uses a verified data foundation.

Acceptance criteria
AC1. Required Assignment 4 data is available in the Statsbook working structure.
AC2. Data required by subsequent analytical stories can be referenced consistently.
AC3. Required location/transaction information is preserved accurately.
AC4. Data preparation does not introduce unsupported observations or assumptions.
AC5. Prepared data supports downstream categorical and numerical analysis.
AC6. Product Owner verification is complete.

A4-02 | Dynamic Categorical Association Analysis
Epic: E1 Excel Statsbook
Team / owner: Defiant / Dume
Story points: 5
Dependencies: A4-01
Traceability: E1 categorical-analysis requirement
Refinement: REFINED / PO APPROVED
Kanban status: DONE / PO ACCEPTED
QA: PASS | Learning Gate: PASS
Open defects: NONE
Accepted baseline: Brittain_DJB_assign4_statsbook Draft 02.xlsx

User story
Approved user story is not recorded on this board. Refer to the refined A4-02 record.

Acceptance criteria
AC1-AC10 are PO approved. Their exact text is not recorded on this board, so it has not been reproduced here. Add the approved AC1-AC10 verbatim.

Scope on Kanban card
Two Categorical Variables worksheet with dynamic row/column selectors, MarketType x StoreFormat default, contingency table, marginal totals, row percentages, dynamic chart, reconciliation QA.

A4-03 | Numerical Analysis and Regression
Epic: E1 Excel Statsbook
Team / owner: Defiant / Dume
Story points: 5 (provisional)
Dependencies: A4-01. Feeds A4-04
Refinement: PROVISIONAL
Kanban status: BACKLOG

User story
As an analyst, I want to perform the required numerical and regression analysis on the prepared Statsbook data so that transaction-volume relationships can be quantified for subsequent forecasting.

Acceptance criteria
AC1. Required numerical variables are analyzed using the Assignment 4 data.
AC2. Required descriptive statistics are calculated from source-supported data.
AC3. Required regression analysis is constructed from the appropriate Assignment 4 variables.
AC4. Regression inputs reference the verified Statsbook data rather than unsupported hard-coded substitutes.
AC5. Regression outputs required for subsequent forecasting are available.
AC6. Calculations can be technically verified.
AC7. Results required downstream can be traced back to the analyzed data.
AC8. No unsupported statistical assumptions or additional models are introduced.

A4-04 | Develop Candidate Transaction-Volume Forecasts
Epic: E1 Excel Statsbook
Team / owner: Defiant / Dume
Story points: 5 (provisional)
Dependencies: A4-02, A4-03. Feeds A4-05
Refinement: PROVISIONAL
Kanban status: BACKLOG

User story
As management, I want candidate transaction-volume forecasts for Golden, Westminster, and Parker so that each proposed investment can be evaluated using analytically supported volume expectations.

Acceptance criteria
AC1. Candidate forecasts are produced for Golden.
AC2. Candidate forecasts are produced for Westminster.
AC3. Candidate forecasts are produced for Parker.
AC4. Forecasts use outputs from the required Assignment 4 analysis.
AC5. Forecast inputs can be traced to the verified data/model.
AC6. Forecast calculations reconcile with their source analytical outputs.
AC7. Forecasts are structured so that they can feed the profitability model.
AC8. No unsupported locations, assumptions, or forecast methods are added.

E2
Profitability Model Integration

A4-05 | Integrate Candidate Forecasts into Profitability Model
Epic: E2 Profitability Model Integration
Team / owner: Defiant / Dume
Story points: 8 (provisional)
Dependencies: A4-04 and prerequisite model baseline. Feeds A4-06
Refinement: PROVISIONAL
Kanban status: BACKLOG

User story
As management, I want the candidate transaction-volume forecasts integrated into the profitability model so that the proposed Golden, Westminster, and Parker investments can be evaluated under the Assignment 4 analytical assumptions.

Acceptance criteria
AC1. Forecast values required by the profitability model are connected to the appropriate model inputs.
AC2. Golden forecast inputs flow through the profitability model.
AC3. Westminster forecast inputs flow through the profitability model.
AC4. Parker forecast inputs flow through the profitability model.
AC5. Existing DCF/profitability calculations remain operational.
AC6. Dynamic uncertainty/random-error behavior remains functional where applicable.
AC7. Scenario/model recalculation propagates through dependent investment calculations.
AC8. Financial outputs reconcile with the underlying forecast/model inputs.
AC9. Existing verified model functionality is not replaced unnecessarily.
AC10. Integration is technically verified before dashboard reporting.

A4-06 | Update Decision Dashboard
Epic: E2 Profitability Model Integration
Team / owner: Defiant / Dume
Story points: 3 (provisional)
Dependencies: A4-05. Feeds A4-09, A4-10
Refinement: PROVISIONAL
Kanban status: BACKLOG

User story
As management, I want the dashboard updated with the Assignment 4 analytical outputs so that the three proposed investments can be reviewed from a concise decision-oriented view.

Acceptance criteria
AC1. Dashboard reflects required Assignment 4 outputs for Golden.
AC2. Dashboard reflects required Assignment 4 outputs for Westminster.
AC3. Dashboard reflects required Assignment 4 outputs for Parker.
AC4. Dashboard values reconcile with the underlying profitability/model calculations.
AC5. Dashboard updates when dependent model calculations change where applicable.
AC6. Presentation remains focused on information required for the Assignment 4 decision.
AC7. Unsupported decorative analysis or additional business assumptions are not introduced.

E3
Python EDA

A4-07 | Perform Required Python Exploratory Data Analysis
Epic: E3 Python EDA
Team / owner: Voyager / Friday
Story points: 5 (provisional)
Dependencies: source data / A4-01. Feeds A4-08
Refinement: PROVISIONAL
Kanban status: BACKLOG

User story
As an analyst, I want to perform the required exploratory data analysis in Python so that the Assignment 4 dataset can be examined using the required development workflow and the resulting evidence can support management interpretation.

Acceptance criteria
AC1. Python analysis uses the Assignment 4 source data.
AC2. Required analysis executes successfully in the designated Python/Jupyter workflow.
AC3. pandas is used where required for data handling/analysis.
AC4. seaborn is used where required for Assignment 4 visualization/EDA.
AC5. Analysis remains limited to the Assignment 4 requirements.
AC6. Generated analytical outputs can be reproduced from the source data.
AC7. Output required for management interpretation is retained.
AC8. No synthetic data is substituted for required source observations.

A4-08 | Interpret Python EDA Results
Epic: E3 Python EDA
Team / owner: Voyager / Friday
Story points: 3 (provisional)
Dependencies: A4-07. Feeds A4-09
Refinement: PROVISIONAL
Kanban status: BACKLOG

User story
As management, I want the required Python EDA results interpreted in business terms so that Python findings can contribute to the Assignment 4 investment analysis.

Acceptance criteria
AC1. Interpretation is supported by actual Python outputs.
AC2. Relevant findings are stated without inventing unsupported causal explanations.
AC3. Interpretation distinguishes observed analytical evidence from business judgment.
AC4. Findings needed by the management-analysis story are clearly available.
AC5. Python interpretation does not contradict the underlying source data.
AC6. Interpretation remains within Assignment 4 scope.

E4
Management Interpretation

A4-09 | Synthesize Management Interpretation
Epic: E4 Management Interpretation
Team / owner: Enterprise / Ultron
Story points: 5 (provisional)
Dependencies: A4-06, A4-08. Feeds A4-10
Refinement: PROVISIONAL
Kanban status: BACKLOG

User story
As FRO management, I want the Excel/modeling and Python evidence synthesized into a clear management interpretation so that expected transaction volumes and investment uncertainty for Golden, Westminster, and Parker can be understood together.

Acceptance criteria
AC1. Interpretation addresses Golden.
AC2. Interpretation addresses Westminster.
AC3. Interpretation addresses Parker.
AC4. Transaction-volume conclusions reconcile with approved analytical outputs.
AC5. Investment/uncertainty discussion reconciles with the profitability model.
AC6. Relevant Python findings are incorporated where required.
AC7. Evidence and management interpretation are distinguishable.
AC8. Unsupported recommendations or assumptions are not introduced.
AC9. Interpretation directly addresses the Assignment 4 business question.

E5
Release and Midterm Readiness

A4-10 | Perform Cross-Deliverable QA and Reconciliation
Epic: E5 Release and Midterm Readiness
Team / owner: Cross-Team. JARVIS QA coordination with Defiant, Voyager, and Enterprise
Story points: 5 (provisional)
Dependencies: A4-06, A4-08, A4-09. Feeds A4-11, A4-12
Refinement: PROVISIONAL
Kanban status: BACKLOG

User story
As the Product Owner, I want the Excel, Python, dashboard, and management-analysis deliverables reconciled so that contradictory numbers, unsupported claims, and requirement gaps are identified before release.

Acceptance criteria
AC1. Excel/model outputs required by the assignment are checked against dependent deliverables.
AC2. Python results used in interpretation match the Python analytical output.
AC3. Dashboard values reconcile with their underlying calculations.
AC4. Management interpretation reconciles with quantitative evidence.
AC5. Golden information is consistent across applicable deliverables.
AC6. Westminster information is consistent across applicable deliverables.
AC7. Parker information is consistent across applicable deliverables.
AC8. Required acceptance criteria for completed stories have objective evidence.
AC9. Detected defects are returned to the appropriate story/workstream rather than silently corrected outside traceability.
AC10. No known material reconciliation defect remains when this story proceeds to PO acceptance.

A4-11 | Complete Midterm Readiness Gate
Epic: E5 Release and Midterm Readiness
Team / owner: Dale + JARVIS
Story points: 3 (provisional)
Dependencies: completed analytical stories and A4-10
Refinement: PROVISIONAL
Kanban status: BACKLOG

User story
As the student/Product Owner, I want to demonstrate understanding of the analytical concepts exercised by Assignment 4 so that assignment execution also prepares me for the BI-5050 midterm.

Acceptance criteria
AC1. Relevant Assignment 4 analytical concepts are identified from completed work.
AC2. Dale can explain the purpose of each relevant analytical step in his own words.
AC3. Dale can explain how required inputs produce the resulting outputs at an appropriate conceptual level.
AC4. Identified misunderstandings are addressed before the readiness gate is accepted.
AC5. Learning validation does not substitute for completing the Assignment 4 deliverables.

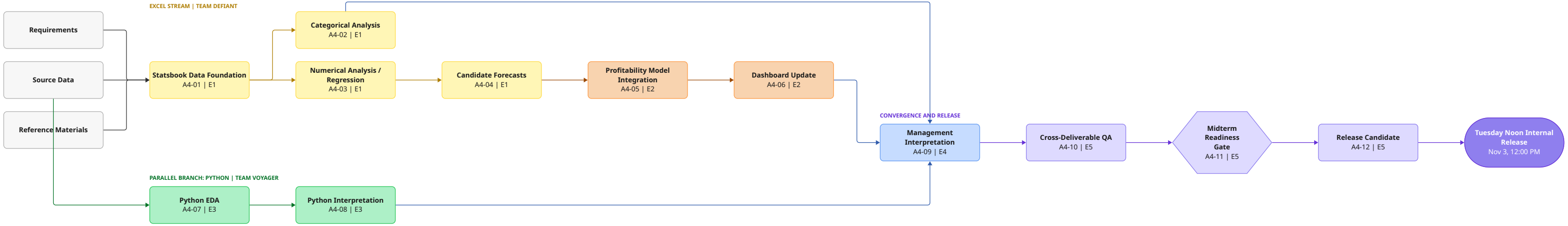
A4-12 | Assemble and Validate Release Candidate
Epic: E5 Release and Midterm Readiness
Team / owner: Cross-Team. Dale / JARVIS with applicable technical team support
Story points: 3 (provisional)
Dependencies: A4-10 and applicable completed deliverable stories. Feeds internal release
Refinement: PROVISIONAL
Kanban status: BACKLOG

User story
As the Product Owner, I want a complete Assignment 4 release candidate assembled and validated so that the required deliverables are ready for the internal release checkpoint and final submission process.

Acceptance criteria
AC1. Required Assignment 4 deliverables are present.
AC2. Only the intended release versions are included.
AC3. Cross-deliverable QA has passed.
AC4. Required numbers and evidence reconcile.
AC5. Known blocking defects are resolved or explicitly returned to development.
AC6. Product Owner performs final acceptance review.
AC7. Recovery/save state is preserved before final release changes.
AC8. Release candidate is ready for the existing Tuesday noon internal release target.

04 Delivery Dependency Map

Read left to right. Each arrow means "must be ready before". Node colors match the Epics frame.



Requirements

Source Data

Reference Materials

EXCEL STREAM | TEAM DEFIANT

Statsbook Data Foundation
A4-01 | E1

Categorical Analysis
A4-02 | E1

Numerical Analysis / Regression
A4-03 | E1

Candidate Forecasts
A4-04 | E1

Profitability Model Integration
A4-05 | E2

Dashboard Update
A4-06 | E2

PARALLEL BRANCH: PYTHON | TEAM VOYAGER

Python EDA
A4-07 | E3

Python Interpretation
A4-08 | E3

CONVERGENCE AND RELEASE

Management Interpretation
A4-09 | E4

Cross-Deliverable QA
A4-10 | E5

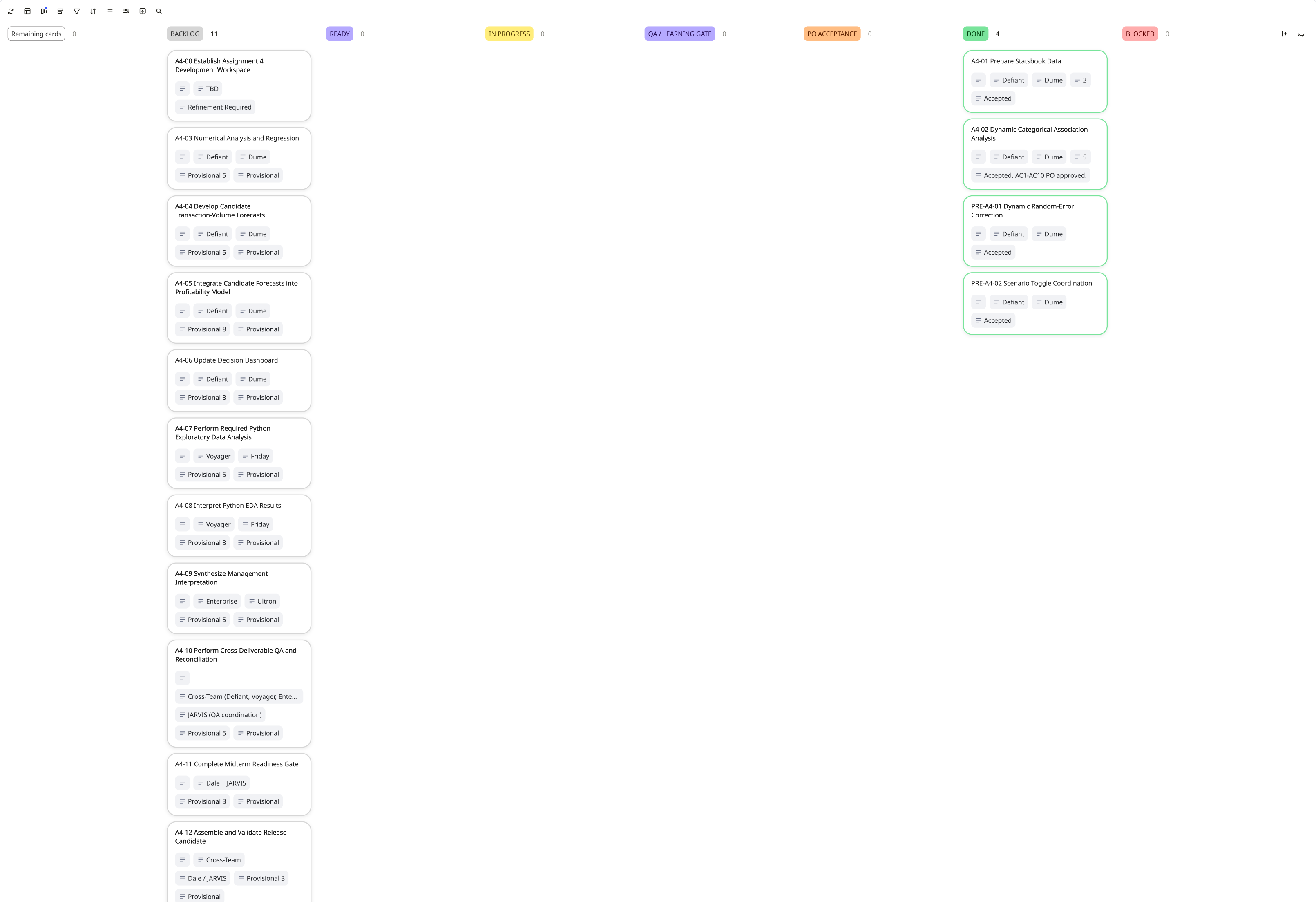
Midterm Readiness Gate
A4-11 | E5

Release Candidate
A4-12 | E5

Tuesday Noon Internal Release
Nov 3, 12:00 PM

05 Story Kanban

Add stories as backlog refinement continues. Drag cards between columns to update status.



Remaining cards 0

BACKLOG 11

- A4-00 Establish Assignment 4 Development Workspace
  - TBD
  - Refinement Required
- A4-03 Numerical Analysis and Regression
  - Defiant
  - Dume
  - Provisional 5
  - Provisional
- A4-04 Develop Candidate Transaction-Volume Forecasts
  - Defiant
  - Dume
  - Provisional 5
  - Provisional
- A4-05 Integrate Candidate Forecasts into Profitability Model
  - Defiant
  - Dume
  - Provisional 8
  - Provisional
- A4-06 Update Decision Dashboard
  - Defiant
  - Dume
  - Provisional 3
  - Provisional
- A4-07 Perform Required Python Exploratory Data Analysis
  - Voyager
  - Friday
  - Provisional 5
  - Provisional
- A4-08 Interpret Python EDA Results
  - Voyager
  - Friday
  - Provisional 3
  - Provisional
- A4-09 Synthesize Management Interpretation
  - Enterprise
  - Ultron
  - Provisional 5
  - Provisional
- A4-10 Perform Cross-Deliverable QA and Reconciliation
  - Cross-Team (Defiant, Voyager, Ente...
  - JARVIS (QA coordination)
  - Provisional 5
  - Provisional
- A4-11 Complete Midterm Readiness Gate
  - Dale + JARVIS
  - Provisional 3
  - Provisional
- A4-12 Assemble and Validate Release Candidate
  - Cross-Team
  - Dale / JARVIS
  - Provisional 3
  - Provisional

READY 0

IN PROGRESS 0

QA / LEARNING GATE 0

PO ACCEPTANCE 0

DONE 4

- A4-01 Prepare Statsbook Data
  - Defiant
  - Dume
  - 2
  - Accepted
- A4-02 Dynamic Categorical Association Analysis
  - Defiant
  - Dume
  - 5
  - Accepted. AC1-AC10 PO approved.
- PRE-A4-01 Dynamic Random-Error Correction
  - Defiant
  - Dume
  - Accepted
- PRE-A4-02 Scenario Toggle Coordination
  - Defiant
  - Dume
  - Accepted

BLOCKED 0

06

Sprint Metrics

Project health. Double-click a value to update it manually.

OVERALL COMPLETION %

--

ACCEPTED STORY POINTS

7 (A4-01 = 2, A4-02 = 5)

TOTAL REFINED STORY POINTS

--

SPRINT COMMITTED POINTS

--

OBSERVED VELOCITY

--

MIDTERM READINESS %

--

DELIVERY CONFIDENCE

--

CURRENT STORY

Backlog refinement / next story pending PO refinement

CURRENT BLOCKER

None

07 Risks / Parking Lot

Add risks on the left, non-blocking ideas on the right. Review the parking lot at Sprint Retrospective.

RISKS

- Tuesday noon internal release target
- Assignment due Tuesday at 5:30 PM
- Assignment and midterm preparation occur concurrently
- RESOLVED / CLOSED by PRE-A4-01: Assignment 3 stored-random-error flagged by instructor as unsuitable for simulation. Dynamic random-error baseline verified.

PARKING LOT

- Future workflow / process improvements
- Optional Miro refinements
- Toolchain improvements that do not block current delivery
- Items identified for Sprint Retrospective

Backlog Traceability

Product Owner rollup of A4 scope by epic. Detailed requirements live on the Product Backlog cards. A4-00 points are TBD and excluded from totals.

ASSIGNMENT 4 ROLLUP
Total A4 stories: 13 (A4-00 to A4-12)
Estimated story points: 52 (A4-00 TBD)
Accepted / DONE story points: 7 (A4-01, A4-02)
By workflow state: BACKLOG 11 | READY 0 | IN PROGRESS 0 | QA / LEARNING GATE 0 | PO ACCEPTANCE 0 | DONE 2 | BLOCKED 0
Overall completion: 7 / 52 points (13.5%), 2 / 13 stories
Current story: Backlog refinement / next story pending PO refinement
Current blocker: None

E1 Excel Statsbook
Stories: 5 (A4-00, A4-01, A4-02, A4-03, A4-04)
Estimated points: 17 (A4-00 TBD) | Accepted / DONE points: 7
BACKLOG 3 | READY 0 | IN PROGRESS 0 | QA / LEARNING GATE 0 | PO ACCEPTANCE 0 | DONE 2 | BLOCKED 0
Complete: 7 / 17 points (41.2%), 2 / 5 stories

E2 Profitability Model Integration
Stories: 2 (A4-05, A4-06)
Estimated points: 11
BACKLOG 2 | READY 0 | IN PROGRESS 0 | QA / LEARNING GATE 0 | PO ACCEPTANCE 0 | DONE 0 | BLOCKED 0
Complete: 0% (0 / 11 points)

E3 Python EDA
Stories: 2 (A4-07, A4-08)
Estimated points: 8
BACKLOG 2 | READY 0 | IN PROGRESS 0 | QA / LEARNING GATE 0 | PO ACCEPTANCE 0 | DONE 0 | BLOCKED 0
Complete: 0% (0 / 8 points)

E4 Management Interpretation
Stories: 1 (A4-09)
Estimated points: 5
BACKLOG 1 | READY 0 | IN PROGRESS 0 | QA / LEARNING GATE 0 | PO ACCEPTANCE 0 | DONE 0 | BLOCKED 0
Complete: 0% (0 / 5 points)

E5 Release and Midterm Readiness
Stories: 3 (A4-10, A4-11, A4-12)
Estimated points: 11
BACKLOG 3 | READY 0 | IN PROGRESS 0 | QA / LEARNING GATE 0 | PO ACCEPTANCE 0 | DONE 0 | BLOCKED 0
Complete: 0% (0 / 11 points)

Historical Prerequisites (not A4 epic scope)

PRE-A4-01 | Dynamic Random-Error Correction
Team / owner: Defiant / Dume | Story points: none recorded (historical, not estimated here) | Dependencies: none within Assignment 4
Refinement: Accepted | Kanban status: DONE / ACCEPTED
Traceability: prerequisite model correction identified before Assignment 4 development

User story: As the Product Owner, I want simulated random error to be calculated dynamically so that recalculation generates new randomized inputs that propagate through the Assignment 4 analytical model.
Acceptance criteria
AC1. Random error is generated dynamically rather than stored as static values.
AC2. Recalculation produces new randomized error values.
AC3. New random values propagate through dependent model calculations.
AC4. Existing model structure required for Assignment 4 remains operational.
AC5. The corrected baseline is verified before dependent Assignment 4 development proceeds.

PRE-A4-02 | Scenario Toggle Coordination
Team / owner: Defiant / Dume | Story points: none recorded (historical, not estimated here) | Dependencies: PRE-A4-01
Refinement: Accepted | Kanban status: DONE / ACCEPTED
Traceability: prerequisite Assignment 4 model stabilization

User story: As the Product Owner, I want scenario controls coordinated with the corrected model baseline so that Assignment 4 analysis can operate from a consistent model state.
Acceptance criteria
AC1. Scenario controls remain compatible with the corrected dynamic-random-error model.
AC2. Scenario changes do not break dependent calculations.
AC3. Baseline behavior is verified before Assignment 4 analytical stories depend on it.